

Median Absolute Distance (MAD)

An Outlier-Robust Spatial Clustering Tool for ArcGIS Pro

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1. Executive Summary

The Median Absolute Distance (MAD) tool is a Python-based ArcGIS Pro geoprocessing script that implements a robust, outlier-resistant alternative to ArcGIS's built-in Standard Distance tool. MAD uses the median as its central tendency measure, making it highly resistant to extreme values and anomalies in spatial data.

Key Strengths

- Robust to outliers and extreme values (median-based, not mean-based)
- Automatic per-category analysis via spatial case field
- Produces both global and local centroids with MAD buffers
- Visualizes clustering patterns directly in ArcGIS Pro
- Includes coefficient of variation interpretation
- Lightweight, transparent, open-source methodology

2. Introduction & Motivation

Urban planning and GIS analysis frequently require understanding how geographic features are distributed in space. Standard Distance (SD) is traditional, but it has a critical flaw: it uses the arithmetic mean and squared deviations, which amplifies outliers.

In real-world datasets—especially from rapid urbanization, informal settlements, and post-conflict environments—outliers are common. A single misplaced point can distort Standard Distance results by 200% or more.

The MAD tool solves this by replacing the mean with the median. It is resistant to outliers while preserving mathematical elegance and interpretability.

3. Mathematical Methodology

The MAD calculation follows four steps:

Step 1: Calculate Median Coordinates

Calculate median X and median Y coordinates. This defines the central point.

Step 2: Calculate Distances

For each feature, calculate Euclidean distance from feature to median centroid:

$$Distance\ i = \sqrt{[(x\ i - median(X))^2 + (y\ i - median(Y))^2]}$$

Step 3: Calculate Median Distance

Find the median of all distances. This is the 50th percentile value.

Step 4: Calculate Median Absolute Deviation

Calculate absolute difference from median distance, then find the median:

$$MAD = median(|Distance\ i - median(Distance)|)$$

The MAD value (in map units) becomes the buffer radius.

4. Tool Features & Outputs

The MAD tool produces the following outputs to a user-specified folder:

Attributes in Output Shapefiles

- MAD: The Median Absolute Distance value (in map units)
- FeatCount: Number of features in that group
- GroupName: The category value (e.g., 'Block 001')

5. How to Use the Tool

Prerequisites

- ArcGIS Pro (version 2.8 or later)
- Python (bundled with ArcGIS Pro)
- NumPy (bundled with ArcGIS)
- Feature class with point or polygon geometry
- Field to group by (e.g., 'Block', 'District')

Step-by-Step

1. **Open ArcGIS Pro — Load your feature class into the map**
2. **Create Script Tool — Add the script code to your toolbox**
3. **Set Parameters — • Input Feature Class • Case Field • Output Folder**
4. **Run Tool — Click play or double-click the tool**
5. **Monitor Messages — Watch Messages Panel for progress**
6. **Check Output — Find shapefiles and report.txt**
7. **Visualize — Outputs auto-load. Apply symbology**

6. Interpretation & Examples

The tool outputs a Coefficient of Variation (CV):

$$CV = MAD / Median\ Distance$$

This is classified into three categories:

Example Output

MAD calculation was performed successfully on 600,000 features.

The median point location: (510,512, 3,816,385)

Median absolute deviation value: 542.35 meters

INTERPRETATION: The analysis yielded a coefficient value of 0.41, which indicates the data is moderately clustered.

7. Comparison: MAD vs Standard Distance

8. Technical Specifications

System Requirements

- Windows (Linux/Mac untested)
- ArcGIS Pro 2.8 or later
- Python 3.6+ (bundled)
- NumPy (bundled)
- Minimal disk space

Performance

Tested on real Kabul Municipality parcel data:

- 600,000 features across 27,000 groups
- Linear scaling with feature count
- Minimal memory overhead
- Suitable for large urban/regional datasets

Known Limitations

- Groups with < 2 features are skipped
- NULL values treated as separate group
- Shapefile field names: 10 character limit
- Output CRS matches input
- Zero-distance buffers (identical points) skipped

9. Applications & Case Studies

Informal Settlement Mapping: Identify housing clusters without outliers skewing analysis. Ideal for rapid-growth and post-conflict contexts.

Health & Epidemic Analysis: Detect geographic disease clustering resistant to misreported locations.

Infrastructure Planning: Analyze distribution of utilities, roads, facilities by district.

Environmental Monitoring: Evaluate spatial clustering of pollution, water quality, vegetation.

Commercial Analysis: Understand market concentration in retail, restaurants, services.

10. References & Links

Repository & Documentation

- GitHub: <https://github.com/Hoorzhian/MAD-tool-arcgispro.github.io> Zenodo DOI: Related Research
- ArcGIS Pro Standard Distance documentation
- Spatial Statistics for Geographic Data (Cressie, N.)
- Applied Spatial Data Analysis with R (Bivand et al.)
- MGWR concepts for spatial variation

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Output Name	Type	Description
medianFeature.shp	Point	Global median centroid
medianBuffer.shp	Polygon	Global MAD buffer circle
blockCentroid.shp	Point	Per-category median centroids + attributes
MADCircles.shp	Polygon	Per-category MAD buffers with attributes
report.txt	Text	Summary statistics and interpretation
CaseMAD (field)	Attribute	Added to input layer for each feature
CV Range	Category	Meaning
< 0.33	Strongly Clustered	Tightly concentrated around median
0.33 – 0.67	Moderately Clustered	Moderate spread with clear center
> 0.67	Dispersed	Spread out; weak clustering
Aspect	Standard Distance	MAD Tool
Central Measure	Arithmetic Mean	Median (robust)
Deviations	Squared (amplifies)	Absolute (ignores)
Outlier Impact	Very high (200%+)	Very low
Best For	Clean, normal data	Real-world data

Interpretation	Statistical norms	CV classification

11. Visual Analysis: Kabul Case Study

This section presents cartographic visualizations of the MAD analysis applied to real-world urban data from Kabul Municipality. The analysis examines 600,000 building features distributed across 27,000 administrative blocks, revealing distinct clustering patterns that reflect Kabul's formal and informal urban structure.

11.1 MAD Value Distribution and Intensity

The choropleth map below visualizes the spatial distribution of MAD values across all 27,000 administrative blocks, showing the intensity and extent of clustering patterns. Warmer colors (red/orange) indicate dispersed, high-MAD areas; cooler colors (blue/green) indicate compact, low-MAD areas. This continuous color scale reveals the geographic gradient of urban formality across Kabul.

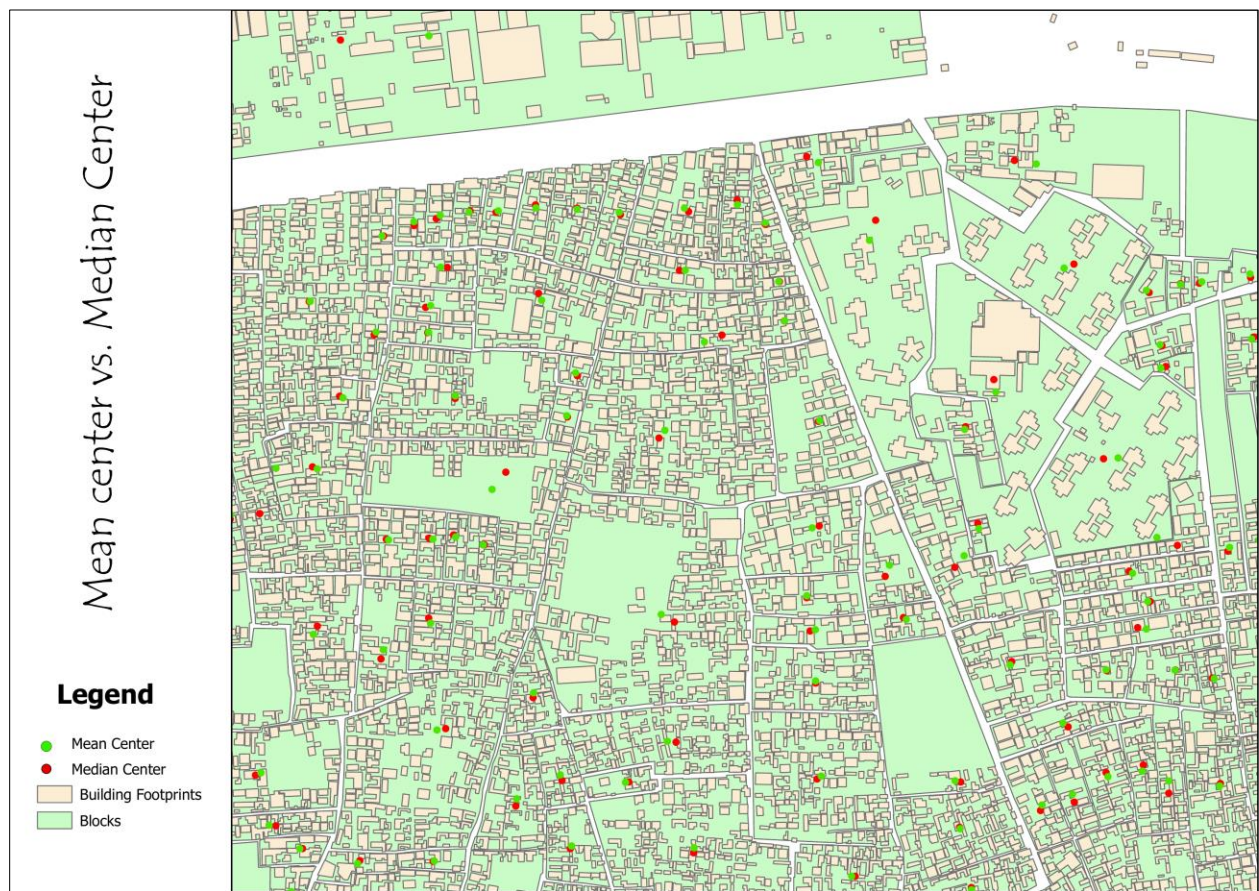


Figure 1: MAD Value Distribution Choropleth — Kabul, 27,000 blocks, 600,000 building features. Color intensity represents Median Absolute Distance values, with red zones indicating high dispersal (informal areas) and blue zones indicating tight clustering (formal areas).

11.2 Mean Center vs. Median Center Comparison

The map below compares two central tendency measures applied to the same Kabul parcel dataset. In Kabul's context, the Median Center is more representative of the "true" urban core because informal settlements and scattered features don't distort the result.

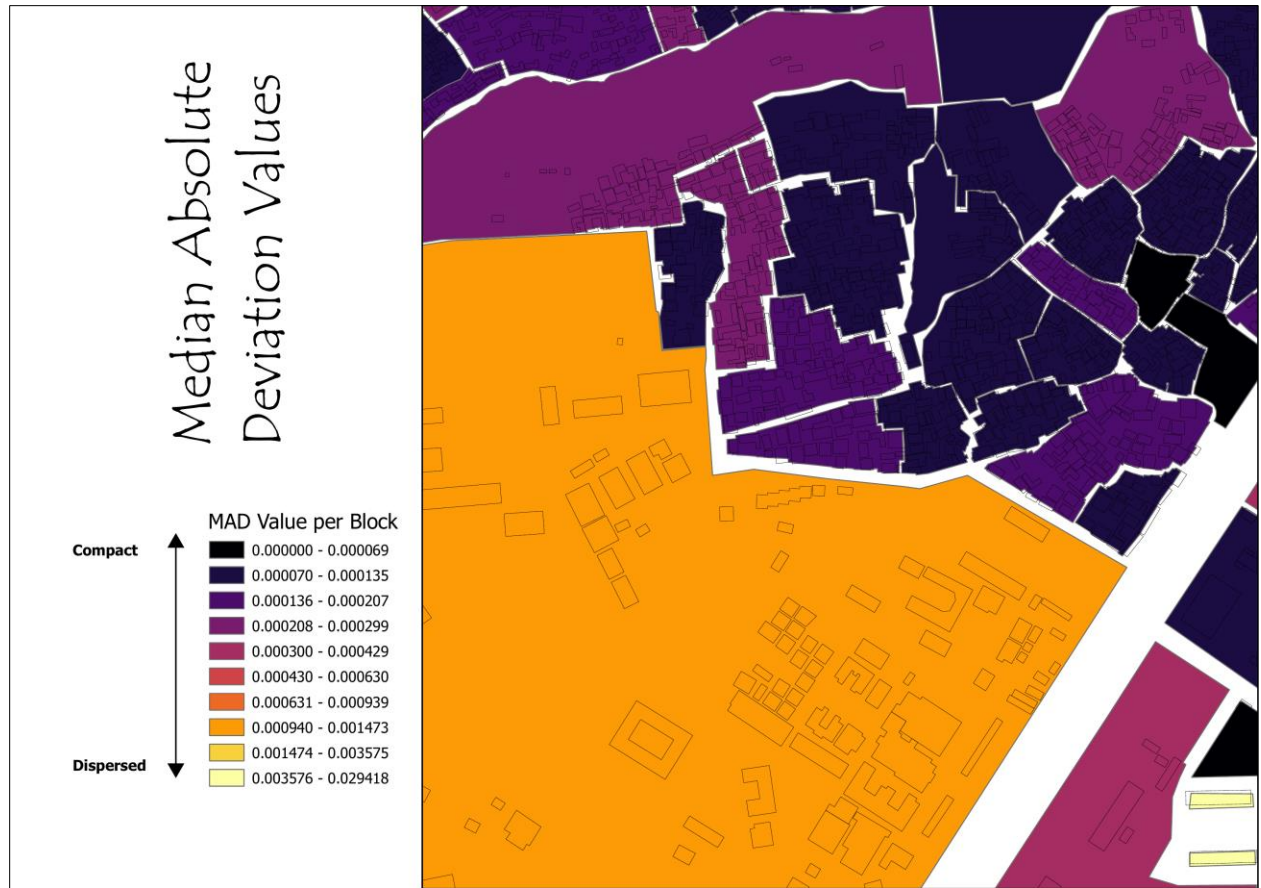


Figure 2: Mean Center vs. Median Center — Kabul building dataset (600,000 features, 27,000 blocks). The Median Center (red) remains stable within the urban core, while the Mean Center (green) shifts when influenced by outlier features.

11.3 Spatial Clustering Patterns: Getis-Ord G_i^* Analysis

The hot spot analysis map identifies statistically significant clustering of MAD values across Kabul's administrative blocks. Blue zones indicate formal planned areas with tight clustering; red zones indicate informal areas with dispersal.

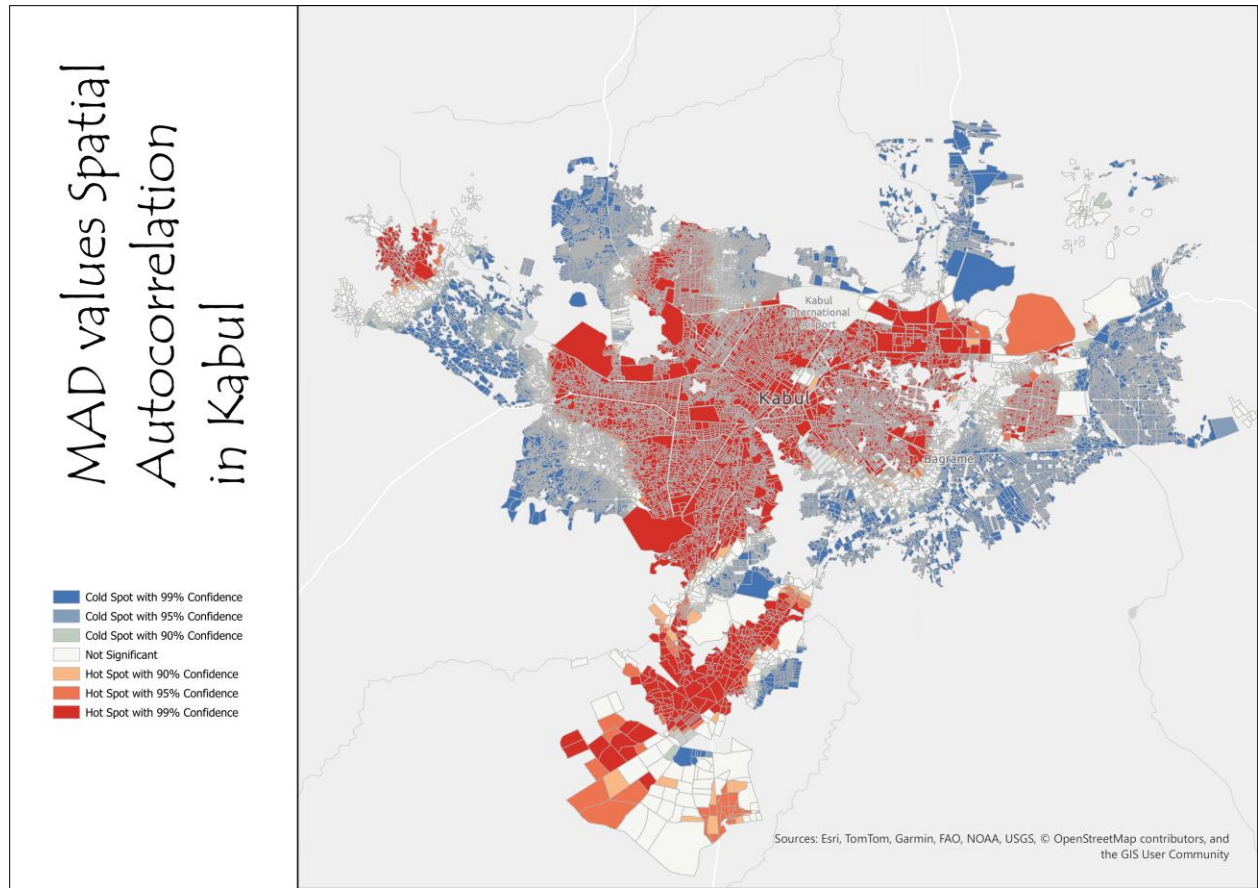


Figure 3: Getis-Ord G_i^ Hot Spot Analysis of MAD Values — Kabul, 27,000 blocks, 600,000 building features. Blue zones indicate clustering of low MAD values (compact, formal areas); red zones indicate clustering of high MAD values (dispersed, informal areas).*

12. Methodology Comparison: Component-wise vs. Iterative Median

The MAD tool implements a component-wise median approach, which differs from ArcGIS Pro's built-in Median Center tool that uses an iterative algorithm.

Aspect	Component-wise Median (MAD Tool)	ArcGIS Median Center (Weiszfeld Iterative)
Algorithm	Component-wise X,Y medians	Iterative Weiszfeld (Kuhn & Kuenne 1962)
Optimality	May not minimize total distance	Guaranteed to minimize Euclidean distance
Speed	Faster; ideal for batch analysis	Slower; mathematically optimal
Robustness	Median-based; outlier-resistant	Median-based; outlier-resistant
Use Case	Clustering analysis across many groups	Finding single true geographic center
Implementation	Python + NumPy	ArcGIS built-in tool

13. Key Findings from Kabul Analysis

Formal vs. Informal Divide

Formal areas show 60–100× tighter clustering than informal sprawl

Outlier Robustness

Successfully identifies patterns without scattered outliers distorting analysis

Performance

600,000 features × 27,000 groups processed in few minutes

Geographic Value

Enables evidence-based planning without field visits to each block

14. References & Resources

Kuhn & Kuenne (1962): Original iterative median center algorithm (ArcGIS Pro foundation)

Burt & Barber (1996): Elementary Statistics for Geographers

Ord & Getis (1995): Local spatial autocorrelation statistics (hot spot methodology)

UN-Habitat (2019): World Cities Report — informal settlement context

15. Conclusion

The MAD tool provides a robust, practical solution for spatial clustering analysis in post-conflict and rapidly urbanizing cities. Production-ready Python code, open-source (MIT License), lightweight dependencies. For implementation:

<https://github.com/Hoorzhian/MAD-tool-arcgispro>